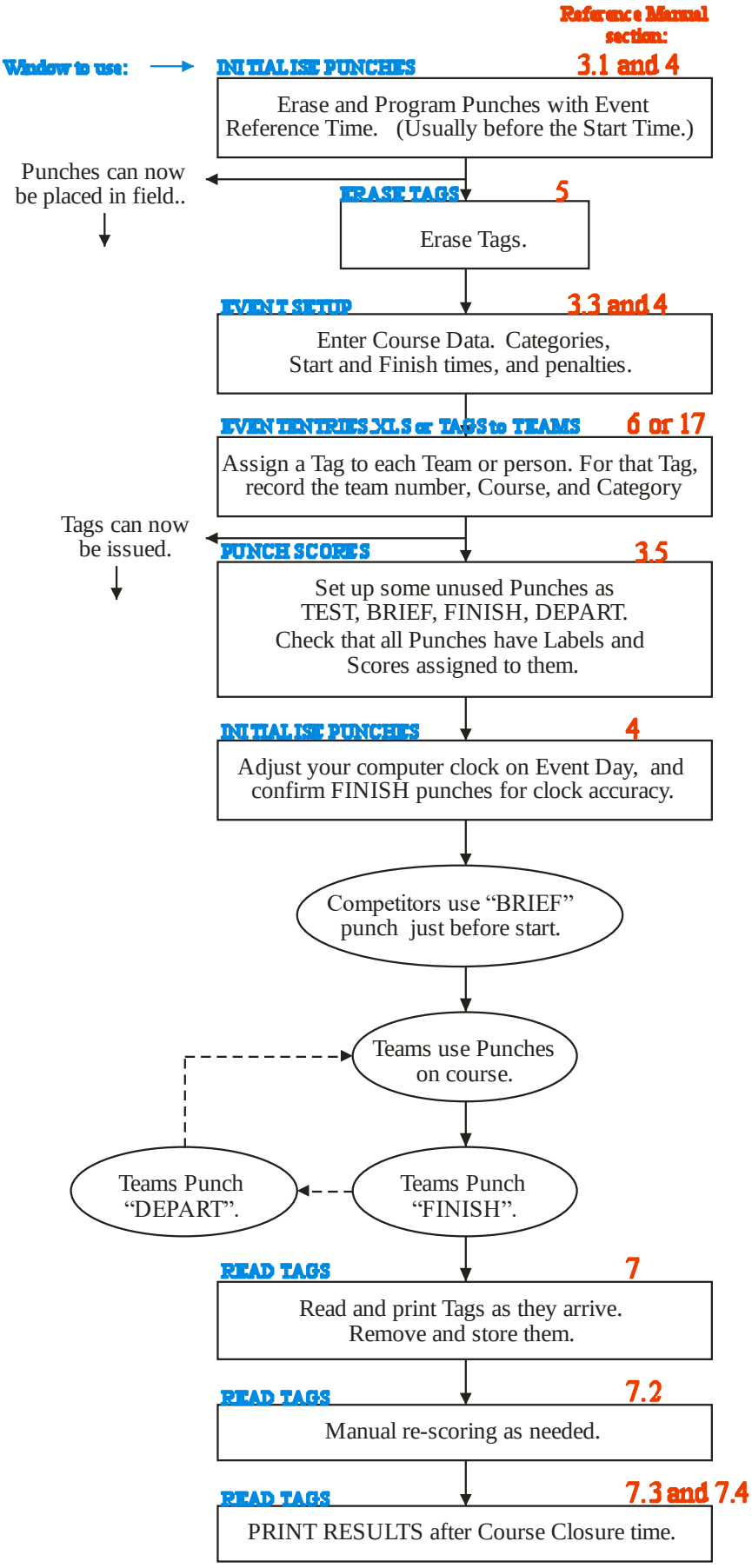
# The NavLight System QUICK REFERNCE.

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# 1 QUICK FLOW DIAGRAM.



## 1.1 What must I decide about my event before I start?

There are several decisions you must make before you set up your event. It is simpler if you make these decisions before you start. You will need:

- An Event name, eg. “Spring Classic”.
- A name for each Course you are having, eg. 6 Hour, 12 Hour, School Championship. Even if you only have one course, you should have a name for it which is different from the Event name. Eg. Course 1 or Main Course.
- The starting and finishing times of each course, and the date of your event.
- The late-point penalty rate for each course, and the time to disqualification. (10 points per minute late, and 30 minutes to disqualification, are standard.)
- What age and gender classes you will have, for example the usual ones are M, W, X, MV, WV, XV, MSV, WSV, XSV, and possibly J. You can also have an Open grade (O) for which everyone competes.
- How many tags you will use: one per team, two per team, or one per person. (Two per team is most common except for championships, which have one tag per person.)
- If you are using more than one tag per team, you must decide NOW what you will do if a team returns with different scores. Will you use the maximum score or the Minimum score (which is normal). This choice can be different for each course.
- An “EVENT REFERENCE TIME”. This should be 2 to 6 hours before your first course starts. It is not critical but should not be changed once the Punches have been programmed.

## 1.2 What to do next.

After you have made your decisions, the sequence is roughly:

- Create your Event folder, with the files you need.
- Set the Event Reference Time using EVENT SETUP. (Look at SPECIFIC HELP from the menu)
- Enter your Course information using EVENT SETUP.
- Assign some punches to be BRIEF and FINISH punches using EVENT SETUP.
- Program the punches, ready for use, using PROGRAM PUNCHES.
- Erase the Tags (and make sure that only YOUR tags exist in Tagnums.txt – best done by deleting Tagnums.txt before you erase tags) from the ERASE TAGS window.
- Get information about teams, competitors, age/gender classes etc. and put it in your spreadsheet. The spreadsheet should be placed in your event folder.
- Use Event Setup to make sure your various Courses and Categories match the ones from your entry form and spreadsheet.
- Assign Tags to teams using your spreadsheet.

Note that many of the Windows have HELP and TOOLS menus which will guide you.

On Event day, ensure that all tags are punched with a BRIEF punch, usually about 15 minutes before the start. Also ensure that the Finish punches have their clocks synchronised by re-confirming them in the PROGRAM PUNCHES window, and correcting any timing errors.

As people finish, remove their tags and read them using the READ TAGS window.

# 2 HANDLING ERRORS AND EXCEPTIONS.

## 2.1 The "WHAT-IF" Section.

1. I didn't use the "BRIEF" punch.
2. I didn't use the "FINISH" punch.
3. I forgot to punch at checkpoints 73 and 29.
4. My Tag stopped working after midnight.
5. Punch 52 wouldn't work but we noted its internal code.
6. We finished at midnight and used the "FINISH" punch but decided to go out again at 7:00 am.
7. The Punches were programmed with a 12:00 reference time, but we need to start the event earlier (or later) because of transport or weather problems.
8. We got a blister and decided to do the 6-hour course instead of the 24-hour course.
9. Some printouts show a Punch as a 3-letter code instead of a Punch number.
10. The punching time of some punches looks crazy, and there is no split-time.
11. We have more than one Tag for our team, and our scores are different.
12. We split up at the event start and joined different teams.
13. We are adding more Punches or Tags to our system.
14. My USB printer doesn't work when the USB-Serial cable is connected.
15. We need to find out who has not come back yet.

### 2.1.1 No "BRIEF" punch.

Read the Tag at the finish in the normal way. The score will credit Punches used more than 30 minutes after the Start Time. Any Punch used before this will show "No BRIEF Punch. No score." beside each Punch in the printout. You can allow these Punches as below:

Re-score the Tag manually. Click "Re-score >>" as described in Sections 7.2 and 7.2.1 of the Reference Manual. Select the Tag to be re-scored by using one of the methods described in the Manual (scroll bar, typing the team number or tag code in the Search box, or re-inserting the Tag.) Click the "Re-score selected Tag -" button or double-click the Tag.
Use the "BRIEF punch..." Menu to choose a line to allow briefing from. The Tag will be re-scored. The Score will display in the Display window.

Click “SAVE” to complete the scoring, and register the change.

Caution: If there is any suspicion that the competitor may have been on the course before the event start, check the new printout (or scored file) to look for the punching times.

If many tags have not been briefed, you may click the “BRIEF punch... Allow Brief at Tag Start” but you will have to re-score>> all the affected tags.

### 2.1.2 No “FINISH” punch.

Read the Tag at the finish in the normal way. The score will show as 0 and there will be a message on the printout saying “No FINISH Punch”. The printout will assume the finish time is one minute after the disqualification time.

Now, re-score the Tag manually. Click “Re-score >>” as described in Section 7.2 of the Manual. Select the Tag to be re-scored by using one of the methods described in the Manual (scroll bar, typing the team number or tag code in the Search box, or re-inserting the Tag.) Click the “Re-score selected Tag -” button or double-click the Tag.

Click “Alter Finish time” and a Red Date-Time Picker will appear. The procedure is described in the Reference Manual, Section 7.2.2

Set the finish time you want to assign. The displayed score will alter as you change the time; instantaneously if you use the arrows, and after a 2-second delay if you type directly into the Date-Time picker. Click SAVE to end the re-scoring of that Tag.

### 2.1.3 Tag Failure.

If a Tag malfunctions while in use, and there is **only one tag per team**, the competitors should prove their visit to the missed checkpoints by noting down the 3-letter Punch ID code inside the clear nozzle of each Punch. Once you can estimate their score, you must create a RawTag file for the malfunctioning Tag from the TAGS TO TEAMS window, as described at the end of Section 17.6 of the Reference Manual. (Go to the TAG to TEAMS window, find the line for the damaged Tag, and click “Make Rawtagfile”.)

Then, you can do a manual rescore of the Tag and give it the manual score you have decided. This procedure is identical to that described in section 7.2.4 of the Manual. Re-score the Tag, enter the score in the box saying “Nil”, then SAVE the change.

If another team member has a tag,

NOTE: If the team has **more than one Tag** and the other tags are OK, a better method is to use the “TEAM Scoring... Lost TAG” menu. A window opens where you enter the team number and click “ACCEPT”. See section 7.2.5 of the Manual.

### 2.1.4 Punch Failure.

The complete failure of a Punch to recognise an inserted Tag will probably affect all teams who visit that Punch after it fails. You will probably get a number of complaints from people as they finish, or when they check their printouts. You can manually rescore every affected Tag, but this can be very time-consuming.

You can click the “Punch failure” button, and enter the number of the missing Punch. A window will appear with the Punch number in it. Clicking the number will toggle it between yellow and grey. If it is yellow when you score a tag, it will be added to the Tag, then turned grey. Click it before scoring each subsequent Tag which claims the missing Punch.

A better solution is to wait until all affected teams have registered their problem, then correct them all at once using the “damaged Punch” procedure in Section 7.2.3 of the Reference Manual. Open the re-score extension, select a single affected Tag and click the “Punch Failure” button. Then type in all the affected team numbers, separated by commas, enter the Punch number and click OK.

It doesn’t matter if the Punch failed partway through the event, or if some team members succeeded in punching and others didn’t. The process searches all the affected Teams and only adds the Punch if not already credited to each Tag.

### 2.1.5 A team Finished early, then resumed scoring.

This is no problem. They must use a Finish punch when they return again; otherwise, their score will stop at the earlier finish. They can return any number of times.

It is recommended that all teams use a FINISH punch when they return to the Administration area, regardless of their future intentions. Create a “DEPART” Punch in the Event Setup - PUNCH Values window, (See Section 3.5.1 of the Reference Manual) and get all teams who return to the course to use this Punch. While not essential, it gives a record of who has gone out again, and also enables an accurate calculation of their “time on course” result. It is *essential* to use a “DEPART” Punch in events which have time-on-course restrictions, such as a 15-in-23 hour course.

### 2.1.6 Altering the Start Time after the Punches are out.

Use the “Event Setup – COURSE data” window to alter start and finish times. See Section 3.3.1 of the Reference Manual. (The Reference Time must not be altered.)

The Punches are loaded with a “Reference Time” which is normally the same as the event start time, but does not have to be. This “Reference Time” is used during Tag Reading to calculate the exact time of punching.

However the “Reference Time” is not used in the calculation of late finishes, time on course etc. These details are derived from the “Course Scoring Options” as described in Section 3.3 of the Reference Manual.

Therefore the event start and finish times, and penalties, for any or all courses can be altered at any time; even after scoring has started. (Although if this happens, you will need to re-score all tags after the time/penalty changes.)

### 2.1.7 Competitor course-changes after Event Start.

Tags are assigned to Team Numbers. If people join or create different teams, use your spreadsheet or the “TAGS to TEAMS” window to assign new team numbers and categories to their existing Tags.

If scoring has already commenced, go to the "READ TAGS" window, where you ***must*** "Delete Scored Records…Summary Files" from the menu bar. Then do Re-Score>>, and rescore ALL tags as described in Section 7.2.8 of the Reference Manual. Note that if manual modifications have been made to the altered tags, you will need to re-do them if there has been a course change.

### 2.1.8 A Punch Number appears as a 3-letter code.

This happens in a printout if the Punch with that code has not had a Number or Score assigned to it. Another window will open, allowing you to enter the label (CP number) and its score value. If you click ACCEPT, the punch will be added to Punchnums.txt.

However it is more probable that you won't know immediately which punch it is. You need to look at your map and the sequence of punches scored by the tag to work out which CP it is supposed to do. Click CANCEL and when you know the details of the CP, re-open the "EVENT SETUP – PUNCH Values" window, locate the Punch ID, then enter the correct details and save the file. (See Section 3.5.2 of the Reference Manual).
You will then probably have to re-score ALL tags.

If this occurs at all, it is likely that the "PunchNums.txt" file has been corrupted. It is a good idea to make a backup of this file after the original setup, to enable a quick re-instatement.

### 2.1.9 Punch times and split-times look stupid.

This will occur if the battery of a Punch has been disconnected at any time since it was programmed. As long as the battery is re-connected (by screwing the clear nozzle into the grey casing of the Punch), the Tag will record its visit. The actual time of punching will be in error, but the correct score will still be counted.

Therefore the short answer is "do nothing". The results will be OK.

The most obvious indication of this error in a printout is the absence of a split-time.

### 2.1.10 Different scores for Tags in the same team.

From the READ TAGS window, click Re-score >>. A blue label saying "Discrepancy" will appear if Tags within the same team have different scores. Click OK and re-score the highlighted Tags at your discretion. See the Reference Manual, Section 6.3.8.

Note that when Results are calculated, the lowest score is normally used if a team has more than one Tag. The option of lowest or highest is controlled in the Event Setup.

### 2.1.11 A Team splits up into other teams at the start.

This is treated as Course changes after event start, and is handled exactly as in section 2.1.7 above. You must re-allocate one of the Tags as a new Team number, with its own course, categories, and names.

### 2.1.12 Integrating new Punches and Tags.

These must have their permanent ID codes stored in PunchNums.txt and TagNums.txt respectively. For Punches, this is done from the "EVENT SETUP—PUNCH values" Window (Section 2.5.3 of the Reference Manual)

Note that any new Punches must have a label (eg. 37) and a score (eg. 30) assigned to them. See Section 3.5.2 of the Reference Manual.

For new Tags, simply erase them for a new event using the "ERASE TAGS" Window, Section 5 of the Reference Manual. This adds them to the TagNums file.

For combining large quantities (whole sets), refer to the Reference Manual.

### 2.1.13 USB Conflict between Printer and USB-Serial cable.

One user found this problem, but others have not. If your USB printer doesn't function properly when the Reader unit is connected using a USB-Serial adapter (which is recommended) then you can work around this problem.

Disconnect the printer, and score tags as they arrive in the usual manner, but with the "PRINT TAGS" checkbox un-ticked. (See Reference Manual section 7.1.1). After several Tags have been scored, unplug the USB-Serial cable from your computer and plug in the printer. Now, click the "Re-score >>" button, select the Tags which have just been read, and manually re-score them with the "PRINT TAGS" checkbox ticked.

Repeat this procedure as often as required.

## 3 BUYING THE WRISTBANDS.

The wristbands are Narrow Vinyl Superbands from Precision Dynamics www.pdcorp.com/crowd-control/superband-classic.html

They may be purchased in NZ from Protec Solutions Ltd. Box 15 259 Wgtn.
120 Tirangi Road Rongotai Wgtn
Phone 04 387 7231 accounts@protecsolutions.co.nz

In Australia, they may be obtained from Belinda Streeting, WIZID PTY LTD.
Ph 02 4351 9566, Fax 02 4351 9566, e-mail sales@wizid.com.au, or website www.wizid.com.au.

These are the stock-code numbers you need. They are usually sold in boxes of 500 at around $180 per box.

| | |
|---|---|
| Stock code PID460-13 | Superband Narrow Blue |
| Stock code PID460-14 | Superband Narrow Yellow |
| Stock code PID460-16 | Superband Narrow Red |
| Stock code PID460-22 | Superband Narrow Green |
| Stock code PID460-29 | Superband Narrow Orange |
| Stock code PID460-33 | Superband Narrow Grape |
| Stock code PID460-37 | Superband Narrow Day Glow Pink |
| Stock code PID460-39 | Superband Narrow Day Glow Green |